Calculate gravitational force

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Calculate a gravitational interaction with consistent units.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 2.*
30 m/s downward.

02. Worked example: explain the model
Opposite attractive 2 N arrows; N·m² divided by m² gives N.

03. Read and interpret the model
The required separation is 4 m, not either half-distance.

04. Apply the physics
r²16 m²; F 1 N; at 8 mF=0.25 N.

05. Explain your reasoning
Small G makes force tiny for modest masses; inability to feel a force does not make it zero.

06. Review: connect an earlier idea *Review: Unit 1, Day 15.*
Only weight 40 N downward from Earth. The hand is no longer in contact; upward velocity is not a force.

07. Exit check
4 N; attractive toward the other body.

08. STRETCH: test the model
At 6 m: 0.5 N. For 18 N, r²=1 m² so r=1 m.

Calculate gravitational force

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Calculate a gravitational interaction with consistent units.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 2.*
v_f=23 m/s down; Δy=26 m down.

02. Worked example: explain the model
Opposite attractive 2 N arrows; N·m² divided by m² gives N.

03. Read and interpret the model
The required separation is 4 m, not either half-distance.

04. Apply the physics
F=0.04002 N≈0.0400 N; at 20 m=0.0100 N.

05. Explain your reasoning
Small G makes force tiny for modest masses; inability to feel a force does not make it zero.

06. Review: connect an earlier idea *Review: Unit 1, Day 15.*
Weight down, normal up, kinetic friction right. No vertical acceleration; rightward horizontal acceleration if friction is the only horizontal force.

07. Exit check
1.6675×10^{-5} N; wrong denominator 2 instead 4 makes answer twice too large.

08. STRETCH: test the model
Product 0.48 N·m²; r=√(0.48/0.03)=4 m.